Properties of Polygons

Exam code: 0607

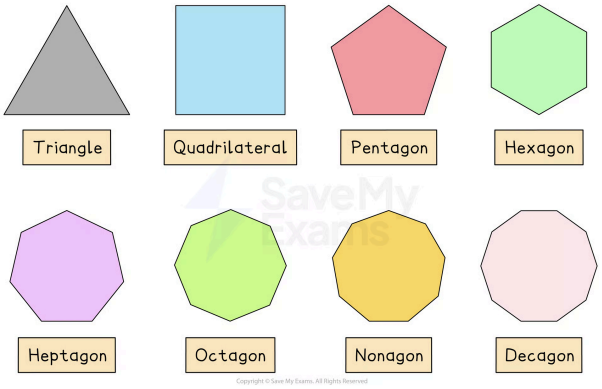
Download PDF

Written by: Jamie Wood
Reviewed by: Dan Finlay
Updated on 18 February 2026

Properties of polygons

What are the names of common 2D shapes?

- You should know the general names of all the 2D **polygons**
  - A **triangle** has 3 sides
  - A **quadrilateral** has 4 sides
  - A **pentagon** has 5 sides
  - A **hexagon** has 6 sides
  - A **heptagon** has 7 sides
  - An **octagon** has 8 sides
  - A **nonagon** has 9 sides
  - A **decagon** has 10 sides
  - A **polygon** is a flat (plane) shape with *n* straight sides
    - A **regular** polygon has all sides the same length and all angles the same size

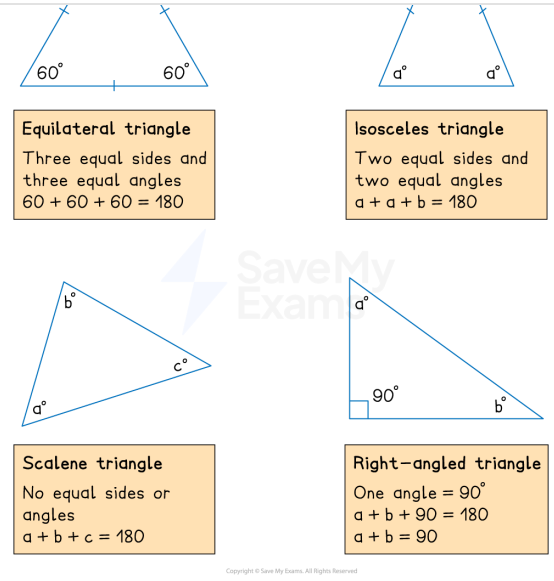


Polygons

What are the names of the different types of triangles?

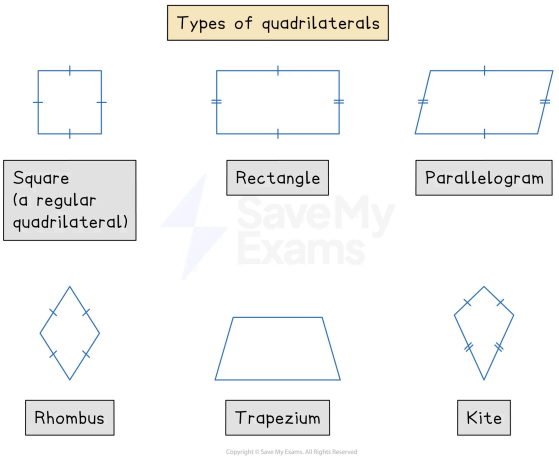
- You should know the names and properties of the different types of **triangles**
  - An **equilateral** triangle has **3 equal sides** and **3 equal angles**
  - An **isosceles** triangle has **2 equal sides** and **2 equal angles**
  - A **right-angled** triangle has **one 90° angle**
  - A **scalene** triangle has **3 sides all of different lengths**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



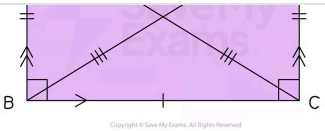
What are the names of the different types of quadrilaterals?

- You should know the names and properties of the different types of **quadrilaterals**
  - These are **squares**, **rectangles**, **parallelograms**, **rhombuses**, **trapeziums** and **kites**



What are the properties of rectangles and squares?

- A **square** is a **rectangle** where all the **sides** have the **same length**
- The table below shows the **properties** of a rectangle and square

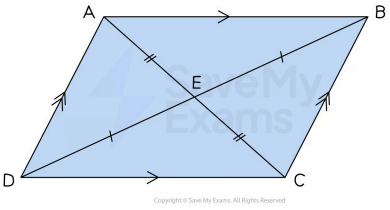


Rectangle

	Rectangle	Square
Sides	• Two pairs of equal, opposite sides • Two pairs of parallel sides	• All sides have the same length • Two pairs of parallel sides
Angles	• All angles are 90°	• All angles are 90°
Diagonals	• Both diagonals bisect each other • Diagonals have the same length	• Both diagonals bisect each other • Diagonals have the same length • Diagonals are perpendicular
Symmetries	• Two lines of symmetry • Order of rotational symmetry is two	• Four lines of symmetry • Order of rotational symmetry is four

What are the properties of parallelograms and rhombuses?

- A **rhombus** is a **parallelogram** where all the **sides** have the **same length**
- The table below shows the **properties** of a parallelogram and rhombus



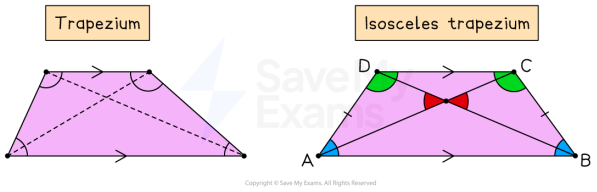
Parallelogram

	Parallelogram	Rhombus

	• Two pairs of parallel sides	parallel sides
Angles	• Two pairs of equal, opposite angles • Each pair of adjacent angles add up 180°	• Two pairs of equal, opposite angles • Each pair of adjacent angles add up 180°
Diagonals	• Both diagonals bisect each other	• Both diagonals bisect each other • Diagonals are perpendicular
Symmetries	• No lines of symmetry • Order of rotational symmetry is two	• Two lines of symmetry • Order of rotational symmetry is two

What are the properties of trapeziums?

- The table below shows the **properties** of a trapezium and an isosceles trapezium

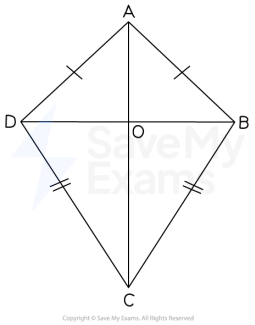


	Trapezium	Isosceles trapezium
Sides	• One pair of parallel sides • The parallel sides have different lengths	• One pair of parallel sides • The parallel sides have different lengths • The non-parallel sides have the same length

	180°	180° • Two pairs of adjacent angles that are equal
Diagonals	• Do not bisect each other	• Do not bisect each other • Diagonals have the same length
Symmetries	• No lines of symmetry • Order of rotational symmetry is one	• One line of symmetry • Order of rotational symmetry is one

What are the properties of kites?

- The table below shows the **properties** of a kite



Kite

	Kite
Sides	• Two pairs of equal, adjacent sides • No parallel sides
Angles	• One pair of equal, opposite angles
Diagonals	• One diagonal bisects the other • Diagonals are perpendicular
Symmetries	• One line of symmetry • Order of rotational symmetry is one

Worked Example

Complete each statement.

Answer:

A quadrilateral with only one pair of parallel sides is called a trapezium.

(ii) A quadrilateral with two pairs of parallel sides is called a .....

Answer:

A quadrilateral with two pairs of parallel sides is called a parallelogram.

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder

Test yourself

Was this revision note helpful? Yes No

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.